Crucis: An Independent Framework for Auditing Causal-Discovery Systems

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Abstract

Causal discovery systems such as Causal-Copilot use language models to help choose an algorithm and review the resulting graph (Wang et al. 2025). However, evaluating these systems solely based on a final benchmark score leaves critical questions unanswered: how much does the result depend on the numerical data versus semantic context (like variable names) or evaluation choices (like scoring rules)? And where exactly in the pipeline does performance change? To systematically answer these questions, we introduce Crucis, an independent framework for auditing externally developed causal discovery systems through controlled comparisons and analysis of recorded stages.

Our recorded audits demonstrate how to isolate these sensitivities across benchmarks, agents, and individual pipeline stages. On the CausalArena semantic scenarios we examined, using just column order to settle uncertain arrow directions raised the Fast Causal Inference (FCI) algorithm's directed F1 by 0.221 relative to a seeded random orientation of uncertain pairs on the same graphs. Furthermore, swapping variable names while holding the data fixed lowered directed F1 by at least 0.10 in 3 of 4 eligible comparisons across entries using different cohorts, scoring rules, and output stages, including an earlier exploratory Causal-Copilot comparison. Finally, in a separate analysis of completed Causal-Copilot runs, we found that the largest paired drop in adjacency F1 occurred during bootstrap refinement, prior to language-model review.

Framework: https://github.com/crucis-lab/crucis
Findings: https://github.com/uzairgheewala/cc-research

1 Introduction 2
2 Prior work 3
3 Data 4
References 6
Appendices A1

1 Introduction

When randomized experiments are impractical, causal discovery systems work to infer causal graphs from observational data under appropriate assumptions about how the data were generated. Increasingly, these pipelines are incorporating large language models (LLMs) to guide the discovery process. For instance, Causal-Copilot uses an LLM to inspect data, select and configure an underlying algorithm, refine the resulting graph, and report the final output (Wang et al. 2025). Other systems have used models to propose structural constraints, suggest a causal order, or directly edit a graph learned from the data (Khatibi et al. 2024; Shen et al. 2025; Vashishtha et al. 2025).

However, as these pipelines grow more complex, standard end-to-end benchmarking obscures exactly why a given system succeeds. A final benchmark score only tells us how closely a graph matches a reference; it leaves open what the system relied on and where in the process the graph changed. Previous work has demonstrated how sensitive these algorithms can be to specific inputs: differences in variable scales can leak causal order in simulated benchmarks (Reisach, Seiler and Weichwald 2021), column order can change the graph a search algorithm returns (Kitson and Constantinou 2024), and LLMs can reconstruct familiar graphs based purely on dataset and variable names alone (Srivastava et al. 2025). These examples motivate the need for independent auditing mechanisms capable of distinguishing which specific input changes affect the reported result and tracing where those changes appear when intermediate outputs are available.

The Crucis Framework. We introduce Crucis, an independent framework for auditing externally developed causal discovery systems. An audit can start with the final output (**L0**) and score it against a stated reference. It can then test how the result responds to a controlled change (**L1**), such as swapping variable names while keeping the numerical values fixed. Where intermediate graphs are available, it can compare successive stages (**L2**) to locate observed transitions between available records and see which edges were added, removed, or reversed. The recorded audits in this report illustrate these approaches, with Causal-Copilot as the worked example for stage tracing.

Crucis currently features components for connecting systems, defining experiments, executing and recording runs under specified conditions, and measuring selected outcomes. The results below come from earlier recorded audits.

Recorded Audits. The audits cover 5 language-model systems and the CausalArena and CausalBench benchmarks. Each example below answers a different question.

Can the scoring rule change the result? On the evaluated CausalArena semantic subset, using column order to settle uncertain directions raised the directed F1 score of Fast Causal Inference (FCI) by 0.221 relative to a seeded random orientation of uncertain pairs on the same graphs. See Appendix A.1 for the scoring comparison.

Can the names change the result? With numerical data fixed, swapping names lowered directed F1 by at least 0.10 in 3 of 4 eligible entries. The entries use different

cohorts, scoring rules, and output stages, including an earlier exploratory Causal-Copilot comparison. Appendix A.2 explains how to read the tally.

Where did the graph change? In the completed Causal-Copilot main-control runs, the largest paired loss in adjacency F1 occurred during bootstrap refinement, before language-model review. Appendix A.3 follows the relevant stages and accounts for crashes.

Contributions. This checkpoint presents Crucis's audit design and shows what it can reveal through earlier recorded audits. Rescoring the same graphs isolates a change in how the benchmark awards credit. Swapping names with the data fixed tests whether the system's output is sensitive to the labels. Comparing successive graphs from the same completed runs shows where performance changed. Together, these comparisons explain why an audit needs to record what a system received, what it returned, and how its output was scored. We document the evidence and limits of each comparison so readers can see which conclusions it supports.

The next work will connect each finding explicitly to the evidence and rule that justifies it, paving the way for verifiable and testable reports of the given systems and of the auditor itself. The longer-term roadmap extends to tracing provenance for individual claims (**L3**) and adversarial tests that search for failure conditions (**L4**).

2 Prior work

What benchmark scores capture. While a reference graph gives us a baseline to score against, the benchmark itself can inadvertently contain clues about the answer. In common simulation settings, simple baselines can recover much of the causal order simply by exploiting patterns in marginal variance or in how well other variables predict each variable (Reisach, Seiler and Weichwald 2021; Reisach et al. 2023). Similarly, even the column order of the input data can change the graph returned by a search algorithm (Kitson and Constantinou 2024). Beyond simulated artefacts, real datasets pose a different problem: the reference graph itself can be highly uncertain, as seen with the widely used Sachs protein-signalling data (Brouillard et al. 2025; Sachs et al. 2005). Newer platforms offer alternative approaches: CausalArena (Li et al. 2026) broadens the range of reference scenarios, while CausalBench (Chevalley et al. 2025; Replogle et al. 2022) evaluates predictions using held-out interventions. These differences matter when interpreting a score: what exactly does it measure, and what reference supports that interpretation?

Language models in the discovery pipeline. This evaluation fragility is compounded when language models enter the pipeline. A variable name can give an LLM useful domain context, but it can also allow the model to reconstruct an answer from familiar dataset and variable names (Kiciman et al. 2023; Srivastava et al. 2025). To test this reliance, frameworks like CARE permute variable names while holding the numerical data fixed,

revealing lower accuracy in those tested settings (Dong et al. 2025). Internal confidence metrics offer little protection here, as language models often report high confidence even on false-positive edges (Kumar et al. 2026). Evaluating LLMs becomes even more complex when they operate as multi-stage agents. Causal-Copilot, for example, places its language-model graph review after bootstrap refinement (Wang et al. 2025). In such multi-stage systems, a change in the final graph raises the critical question of exactly where the change first appeared. Because these pipelines obscure the origin of causal decisions, Zheng et al. call for explicit audit trails that distinguish the data, assumptions, algorithmic outputs, and agent recommendations (Zheng et al. 2026).

Precedents for independent audits. Several frameworks offer precedents for isolating these vulnerabilities. Heuresis traces agent edits within its own evaluation harness and specifically checks the instruments that produce those traces (Ma, Ding and Wang 2026). Other frameworks target distinct failure modes: ACDB separates recovery tasks and tests a structure-blind baseline, TCD-Arena varies time-series assumptions, and Causal-Audit uses diagnostics and estimated risk to recommend a method or abstain (Deb, Shah and Krishnan 2026; Stein et al. 2026; Ruiz et al. 2026). However, for Crucis, the outstanding design question is which of these comparisons can be supported by the records supplied by externally developed systems with different levels of visibility, and how to explicitly report what remains unresolved when those records are missing. To probe such systems, we turn to metamorphic testing, which checks whether outputs follow an expected relationship under a controlled input change (Chen, Cheung and Yiu 1998; Segura et al. 2016; Ribeiro et al. 2020). We combine this with provenance standards that explicitly record where the resulting evidence came from (Moreau and Missier 2013). Together, these disciplines motivate an audit record that transparently identifies the control, the scoring rule, and the exact evidence behind each reported comparison.

3 Data

To execute these comparisons, the audit draws on a suite built from CausalArena's semantic scenarios, named datasets provided by Causal-Copilot, and CausalBench's perturbation data. Table 1 summarizes what each audit uses as a reference, ranging from known simulated graphs to held-out interventions. The checkpoint records source hashes so the exact evidence used can be identified, with exact dataset dimensions, edge counts, and licences deferred to Appendix A.4.

Simulated graphs with exact references. CausalArena provides simulated structural causal models where variables carry names from written scenarios (Li et al. 2026). Unlike real-world datasets, the underlying structure of these simulations is perfectly known, allowing scores to measure accuracy against the generating graph. To support our name-swapping controls, we change the labels while preserving numerical values across 3 distinct contexts: real names, neutral names, and mismatched names. A separate layout audit uses

Table 1: Selected inputs by audit. The evaluation reference indicates what a score actually measures: accuracy against a simulated graph, agreement with an expert reference, or prediction of held-out interventions. Exact dimensions, edge counts, and licences are deferred to Appendix A.4.

Group	Dataset	Evaluation reference
CausalArena	Semantic scenarios	exact
Named	Sachs	contested
	Abalone	provenance
	Auto MPG	mismatched
	Concrete strength	provenance
	DWD climate	provenance
CausalBench	K562	held-out interventions
	RPE1	held-out interventions

CausalArena's public files (semantic, formula, and synthetic) to explicitly examine the effects of column order and scoring rules.

Named datasets with qualified references. To audit real-world datasets, we utilize the exact copies of Sachs and other named tables that ship directly with Causal-Copilot (Wang et al. 2025; Sachs et al. 2005). Crucially, the reference graphs for these datasets require qualification: some have documented provenance without an independently established ground truth, while others differ across published versions. For example, the selected Sachs reference graph differs significantly from its bnlearn and CausalArena counterparts. Similarly, Auto MPG (Quinlan 1993) uses a reference that mismatches CausalArena's version. Consequently, a score on these datasets strictly measures agreement with one selected reference.

Held-out interventions without reference graphs. For evaluations lacking a predefined reference graph, CausalBench uses CRISPRi Perturb-seq screens in K562 and RPE1 cell lines (Chevalley et al. 2025; Replogle et al. 2022). This dataset provides an alternative evaluation regime: rather than comparing predictions against a static graph, it scores them directly against held-out interventions. This lets us examine what the benchmark's metrics reward when a reference graph is unavailable.

Available records and scoped limits. Beyond static datasets, stage comparisons rely on the intermediate records captured during historical runs. The instrumented Causal-Copilot series logs code identity, API requests, intermediate stage graphs, and refinement steps, which directly support our pipeline tracing. By contrast, the name-control results for other systems use a different semantic cohort and rely on their own output contracts, meaning they cannot be paired scenario-for-scenario with the Causal-Copilot traces. Finally, the

limits of these historical runs must be explicitly bounded: the semantic scenarios contain simulation artefacts, each table arm was run once within each Causal-Copilot series, and the Causal-Copilot collection used a single model family. The Crucis protocol records these constraints as the declared evaluation regime, explicitly stating the limits on interpreting these historical runs.

Code

The Crucis repository contains the reusable auditing framework and its documentation. The research repository holds the paper, supporting analyses, and stage-specific attribution records. The number build records source-base commit `aef649241abe`, while the underlying scientific evidence is bound by frozen manifests, file hashes, and separately preserved CausalBench metadata (Appendix A.4). A separate companion reproduction recomputes the recorded Causal-Copilot run comparison from retained graphs with a single command and no API keys. Finally, Appendix A.4 explicitly maps every reported value to its corresponding result file, diagnostic snapshot, or benchmark metadata.

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Appendices

A.1 Scoring the same graphs A1
A.2 Changing names while holding the data fixed A1
A.3 Locating a change inside a pipeline A2
A.4 Where every number comes from A3

Use this appendix to follow an audit from its question to the evidence behind the answer. Start with scoring (Appendix A.1), names (Appendix A.2), or pipeline stages (Appendix A.3). Appendix A.4 is the lookup for exact values, dataset provenance, licences, and their sources.

A.1 Scoring the same graphs

CausalArena's public inventory comprises 250 semantic, 250 formula, and 100 synthetic files. The released files list variables in causal order, and the audited scoring rule uses column position to settle arrow directions left uncertain by the discovery algorithm. On the evaluated semantic subset, Fast Causal Inference (FCI) gains 0.221 directed F1 under that column-order rule relative to a seeded random orientation of uncertain pairs per file on the same graphs ($F1_random$). Also on the semantic subset, the published PC row is 0.333, while a separate PC maxP variant scores 0.463 on shuffled columns under seeded random orientation of uncertain pairs. Because those two PC rows differ in algorithm configuration as well as column order and scoring rule, their difference does not isolate layout and scoring alone.

The separate CausalBench audit results show what the benchmark's metrics reward when a reference graph is unavailable. All 4 published MeanDifference rows reproduce within recorded tolerances. At the smaller evaluated edge budget, MeanDifference reaches 0.90–0.95 of a test-informed reference that ranks pairs by knockdown effects measured on the test split. Its false omission rate stays within the audit's declared margin of the matched random-network mean in 3 of the 4 evaluated cell-line and edge-budget combinations, noting that the random-network comparison was unavailable in the remaining combination.

A.2 Changing names while holding the data fixed

The cross-system name-control tally (audit `AB-06`) pairs graphs produced from the same numerical table under real and swapped variable names. The tally applies a 5-pair entry

floor to each system's declared output stage. PyWhy-LLM (PyWhy 2026) falls below that floor.

Swapping names lowers directed F1 by at least 0.10 in 3 of 4 eligible entries. The drops are 0.251 for causaliq-knowledge (CausalIQ 2026), 0.167 for Causal-Copilot, and 0.122 for MATMCD. The pipeline of Vashishtha et al. (2025) shows a drop of 0.020 under its tested conditions. The tally describes sensitivity under each entry's measurement: the earlier exploratory Causal-Copilot comparison scores committed marks in the graph handed to cycle removal, whereas the other agents use final outputs and seeded random orientation of uncertain pairs on a semantic cohort with no shared scenarios. The tally therefore cannot rank these systems under common conditions.

Which reading do the retained changes follow? Inside Causal-Copilot, our edge-level label-following diagnostic (`BND-08`) examines how completed outputs respond to label changes across three elements:

1. **Selection:** We count distinct (scenario, directed-edge) pairs retained in completed swapped-name outputs, where the pair state differs from the bootstrap graph, matches its post-review state at final output, and is adjacent in exactly one reference reading (either the true data graph or the mismatched label graph; pairs adjacent in both or neither are outside this denominator).
2. **Count:** Among these 101 informative pairs, 81 are adjacent under the label reading and 20 under the data reading. This classification tests adjacency, not agreement of edge direction.
3. **Interpretation limits:** The recorded decision rule comes from `BND-08`'s pre-run specification (with its bytes bound by the preserved manifest, which does not establish when the rule became public). However, that specification describes a pre-cycle endpoint while the recorded calculation consumes completed final outputs. Failed swapped runs cannot enter this completed-output calculation; edges from the same scenario are dependent, and the counts remain descriptive, without claiming confirmation of the specified endpoint.

A.3 Locating a change inside a pipeline

In paired stage diagnostics across the 22 completed Causal-Copilot main-control runs, the largest observed stage loss in adjacency F1 occurs during bootstrap refinement, before language-model review, as located in Figure A 1. Tracking the same completed runs keeps the run set consistent across stage comparisons: across all 30 initial tables, the mean initial adjacency F1 is 0.584, whereas the final mean is 0.321 among the 22 completions. Those initial and final means describe different populations, and a real-arm-only effect answers a separate question not substituted here. Finally, the completed-run comparison leaves out

a separate failure mode: all 8 runs that reached cycle removal crashed there, with logs recording a key error in 8 of those entrants.



Figure A 1: The bootstrap transition had the largest paired drop in adjacency F1 among the completed Causal-Copilot main-control runs. The shaded stage locates that transition before language-model review.

A separate comparison against a data-only graph. When internal stage logs are not decomposed step-by-step, an audit can compare an agent's final LLM-refined output against its own data-only baseline. In the real-name MATMCD runs, pooled precision among committed claims on the released intervention targets is lower for the final output than for the data-only variant on the point estimate (0.6855 against 0.7199); the interval for the difference includes zero. The pre-registered point rule applies to the estimate.

A.4 Where every number comes from

Generated by `docs/paper/q1/build_numbers.py` from `numbers.json` at build-base commit `aef649241abe`. Paths are relative to the root of `cc-research`. Selectors extend JSON pointers: `*` takes every child and `[k=v]` selects the first matching list item. Source flags identify inputs outside the tracked build base or with differences from it. The CausalBench filtering metadata are separately preserved and hashed. The table locates evidence for each value; the companion replay covers only its stated comparison.

Dataset provenance, reference counts, and licences. The evaluation suite draws on the following sources and terms:

CausalArena semantic suite: Released under CC BY 4.0 on Hugging Face (Li et al. 2026). Our test split uses 10 semantic scenarios from the first replicate of the observational-only release (1,000 rows each, 20–25 variables, 35–58 reference edges). Each scenario is written 3 times (real, neutral, and mismatched names) for 30 tables total, with columns shuffled using a fixed seed and opaque file paths.

Sachs: Taken from the copies shipped with Causal-Copilot (Wang et al. 2025; Sachs et al. 2005) (853 rows, 11 variables) across 4 arms: the original, a copy under another path, neutral names, and swapped names. The repository's reference graph has 19 edges, whereas the bnlearn version has 17 (sharing 16 adjacencies with 1 reversed) and CausalArena's version has 20 edges. Checked sources did not identify a licence for the Sachs data.

Abalone, Auto MPG, and Concrete strength: Shipped with Causal-Copilot (4 non-Sachs named tables total, each evaluated across 3 arms: real, neutral, and swapped) and originating from the UCI repository under CC BY 4.0. Abalone (Nash et al. 1994) has 4,177 rows, 8 variables, and 7 expert reference edges (provenance tier). Concrete compressive strength (Yeh 1998) has 1,030 rows, 9 variables, and 8 expert reference edges (provenance tier). Auto MPG (Quinlan 1993) has 392 rows, 5 variables, and 5 reference edges, which differs from CausalArena's version (398 rows, 4 variables, 3 edges; mismatched tier).

DWD climate: Shipped with Causal-Copilot (349 rows, 5 variables, 4 expert reference edges, 3 arms; provenance tier). The copy's lineage is inferred from its match to the Deutscher Wetterdienst pairs in the Tübingen cause-effect benchmark (Mooij et al. 2016); DWD open-data terms state CC BY 4.0.

CausalBench perturbation screens: Uses the CRISPRi Perturb-seq screens of Replogle et al. (2022) released under CC BY 4.0, with CausalBench benchmark code under Apache 2.0 (Chevalley et al. 2025). Retained filtering metadata record 162,751 cells (10,691 unperturbed controls) and 622 perturbed genes in K562, and 162,733 cells (11,485 controls) and 383 genes in RPE1.

Recorded Causal-Copilot runs: Executed unmodified (upstream commit `a31aa7c`) with gpt-4o-2024-08-06, comprising 30 runs on the semantic suite in `BND-08` and 30 main-control runs on the same tables (API cost $12.57).

Key	Value	Source file and JSON pointer (cc-research)
`cc.arms`	30	`analysis/results/fork/bld06_live_compare.json /arms`
`cc.completed`	22	`analysis/results/fork/bld06_live_compare.json /completed/main`
`cc.s0-adj`	0.584	`analysis/results/fork/bld06_live_compare.json /means/main/S0_adj_f1`
`cc.final-adj`	0.321	`analysis/results/fork/bld06_live_compare.json /means/main/final_adj_f1`
`cc.usd`	$12.57	`analysis/results/fork/bld06_live_compare.json /usd/main`
`cc.cyc.entered`	8	`analysis/results/fork/bld06_cycles.json /arms/main/runs_entering_cycle_step`
`cc.cyc.keyerr`	8	`analysis/results/fork/bld06_cycles.json /arms/main/cycle_step_keyerrors`
`bnd08.runs`	30	len of: `analysis/results/stability_s2/bnd08_summary.json /runs`
`bnd08.m3.labels`	81	`analysis/results/stability_s2/bnd08_summary.json /M3/labels_only`
`bnd08.m3.data`	20	`analysis/results/stability_s2/bnd08_summary.json /M3/data_only`
`bnd08.m3.n`	101	`analysis/results/stability_s2/bnd08_summary.json /M3/n`

Key	Value	Source file and JSON pointer (cc-research)
bnd08.model	gpt-4o-2024-08-06	analysis/results/stability_s2/bnd08_summary.json /snapshots/0
ab06.agents	4	len of: analysis/results/ab06/ab06.json /agents
ab06.cc-entered	1	count_true of: analysis/results/ab06/ab06.json /causal_copilot/*/counted_in_P9
ab06.systems	5	add of: key ab06.agents; key ab06.cc-entered
ab06.p9.entered	4	len of: analysis/results/ab06/ab06.json /P9/systems_entered
ab06.p9.losing	3	len of: analysis/results/ab06/ab06.json /P9/losing_ge_0.10
ab06.p9.min-scen	5	analysis/results/ab06/ab06.json /P9/min_paired_scenarios
ab06.p9.threshold	0.10	plans/ab06_prereg.json /predictions/P9/supported (pattern lose ⩾ ([0-9.]+))
ab06.causaliq.drop	0.251	analysis/results/ab06/ab06.json /agents/causaliq/name_reliance_real_minus_swapped/mean
ab06.cc.drop	0.167	analysis/results/ab06/ab06.json /causal_copilot/causal_copilot_gpt4o/name_reliance_real_minus_swapped/mean
ab06.matmcd.drop	0.122	analysis/results/ab06/ab06.json /agents/matmcd/name_reliance_real_minus_swapped/mean
ab06.vashishtha.drop	0.020	analysis/results/ab06/ab06.json /agents/vashishtha/name_reliance_real_minus_swapped/mean
ab06.matmcd.dv1	0.6855	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_final
ab06.matmcd.dv1-data	0.7199	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_data_only
ab01.pc.published	0.333	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, default rule);family=semantic]/published
ab01.pcmaxp.controlled	0.463	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, maxP);family=semantic]/ours_shuffled_expectation
ab01.fci.authors	0.578	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_authors_rule
ab01.fci.expect	0.357	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_expectation
ab01.fci.gain	0.221	sub of: key ab01.fci.authors; key ab01.fci.expect

Key	Value	Source file and JSON pointer (cc-research)
ab01.files.semantic	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=semantic]/files
ab01.files.formula	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=formula]/files
ab01.files.synthetic	100	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=synthetic]/files
ben13.reproduced	4	analysis/results/ben13/summary.json / reproduced_rows
ben13.cells	4	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/placement_share_wasserstein
ben13.for.random-level	3	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/?placement_share_for_note
ben13.echo.min	0.90	min of: analysis/results/ben13/summary.json /predictions/P3/values/*
ben13.echo.max	0.95	max of: analysis/results/ben13/summary.json /predictions/P3/values/*
s2.families	10	len of: datasets/suite_q1s2/MAPPING.json /families
s2.tables	30	len of: datasets/suite_q1s2/MAPPING.json /arms
s2.arms-per	3	div of: key s2.tables; key s2.families
s2.rows	1,000	max of: datasets/suite_q1s2/MAPPING.json /families/*/rows
s2.vars.min	20	min of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.vars.max	25	max of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.edges.min	35	min of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s2.edges.max	58	max of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s1.sachs.rows	853	datasets/MAPPING.json /source/rows
s1.sachs.vars	11	len of: datasets/MAPPING.json /source/columns
s1.sachs.edges	19	len of: datasets/MAPPING.json /source/reference_edges
s1.sachs.arms	4	len of: datasets/MAPPING.json /arms
s1.sachs.bnlearn	17	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/edges_b
s1.sachs.shared	16	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/shared_adjacencies

Key	Value	Source file and JSON pointer (cc-research)
s1.sachs.reversed	1	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/reversed
s1.sachs.arena	20	analysis/results/reference_reliability/reliability.json /families/sachs/causalarena_version/edges
s1.abalone.rows	4,177	datasets/suite_b1/MAPPING.json /families/abalone/rows
s1.abalone.vars	8	len of: datasets/suite_b1/MAPPING.json /families/abalone/columns
s1.abalone.edges	7	len of: datasets/suite_b1/MAPPING.json /families/abalone/reference_edges
s1.autompg.rows	392	datasets/suite_b1/MAPPING.json /families/autompg/rows
s1.autompg.vars	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/columns
s1.autompg.edges	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/reference_edges
s1.concrete.rows	1,030	datasets/suite_b1/MAPPING.json /families/concrete/rows
s1.concrete.vars	9	len of: datasets/suite_b1/MAPPING.json /families/concrete/columns
s1.concrete.edges	8	len of: datasets/suite_b1/MAPPING.json /families/concrete/reference_edges
s1.dwd.rows	349	datasets/suite_b1/MAPPING.json /families/dwd/rows
s1.dwd.vars	5	len of: datasets/suite_b1/MAPPING.json /families/dwd/columns
s1.dwd.edges	4	len of: datasets/suite_b1/MAPPING.json /families/dwd/reference_edges
s1.autompg.arena.vars	4	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/0
s1.autompg.arena.edges	3	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/1
s1.autompg.arena.rows	398	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/2
s1.named.tables	12	len of: datasets/suite_b1/MAPPING.json /arms
s1.named	4	len of: datasets/suite_b1/MAPPING.json /families
s1.named.arms-per	3	div of: key s1.named.tables; key s1.named
cb.k562.cells	162,751	external/causalbench/dataset_k562_filtered.npz.json /cells_out *(excluded)*
cb.k562.genes	622	external/causalbench/dataset_k562_filtered.npz.json /genes_out *(excluded)*

Key	Value	Source file and JSON pointer (cc-research)
cb.k562.controls	10,691	external/causalbench/dataset_k562_filtered.npz.json /controls *(excluded)*
cb.rpe1.cells	162,733	external/causalbench/dataset_rpe1_filtered.npz.json /cells_out *(excluded)*
cb.rpe1.genes	383	external/causalbench/dataset_rpe1_filtered.npz.json /genes_out *(excluded)*
cb.rpe1.controls	11,485	external/causalbench/dataset_rpe1_filtered.npz.json /controls *(excluded)*